Applied Soft Computing Journal 164 (2024) 112000

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Underwater image enhancement via cross-wise transformer network focusing on pre-post differences[☆]

Zhixiong Huang[a], Jinjiang Li[b], Xinying Wang[a], Zhen Hua[c], Shenglan Liu[e], Lin Feng[d,e,*]

[a] School of Computer Science and Technology, Dalian University of Technology, Dalian 116081, Liaoning, China
[b] School of Computer Science and Technology, Shandong Technology and Business University, Yantai 264005, Shandong, China
[c] School of Information and Electronic Engineering, Shandong Technology and Business University, Yantai 264005, Shandong, China
[d] School of Information and Communication Engineering, Dalian Minzu University, Dalian 116600, Liaoning, China
[e] School of Innovation Experiment, Dalian University of Technology, Dalian 116081, Liaoning, China

ARTICLE INFO

Keywords:
Underwater image enhancement
Cross-wise transformer
Attention mechanism
Multi-scale feature

ABSTRACT

The underwater images often suffer from color deviations and blurred details. To address these issues, many methods employ networks with an encoder/decoder structure to enhance the images. However, the direct skip connection overlooks the differences between pre- and post-features, and deep network learning introduces information loss. This paper presents an underwater image enhancement network that focuses on pre-post differences. The network utilizes a multi-scale input and output framework to facilitate the underwater image enhancement process. A novel cross-wise transformer module (CTM) is introduced to guide the interactive learning of features from different periods, thereby enhancing the emphasis on detail-degraded regions. To compensate for the information loss within the deep network, a feature supplement module (FSM) is devised for each learning stage. FSM merges the multi-scale input features, effectively enhancing the visibility of underwater images. Experimental results across several datasets demonstrate that the integrated modules yield significant enhancements in network performance. The proposed network exhibits outstanding performance in both visual comparisons and quantitative metrics. Furthermore, the network also exhibits good adaptability in additional visual tasks without the need for parameter tuning. Code and models are released in https://github.com/WindySprint/CTM.

1. Introduction

Given the intricacy and hazards of the underwater environment, autonomous underwater vehicles (AUVs) guided by underwater sensing assume a critical function in real-world underwater operations, encompassing tasks such as resource exploration, underwater salvage, and biological tracking. To facilitate effective visual sensing, underwater images possessing accurate color representation and sharpness hold immense importance [1]. However, underwater images are often afflicted by issues of color deviation and blurred details [2].

In intricate underwater environments, the degradation process of underwater images is primarily influenced by two main factors [3]. Firstly, the presence of abundant suspended particles (like dissolved organic matter, microplankton, etc.) in the underwater environment initiates a series of transformations in the propagation direction of light. These particles introduce random alterations in the direction of light propagation, causing the light captured by the camera to encompass elements of direct reflection, forward scattering, and backward scattering. These phenomena of forward and backward scattering contribute to the blurring of content and details within underwater images, respectively [4]. The left side of Fig. 1 shows the impact of particles on the propagation of underwater light. Secondly, varying wavelengths of light have distinct attenuation rates in the underwater medium. Longer wavelengths undergo greater absorption effects. Consequently, red light, with the longest wavelength, experiences significant loss as depth increases. On the other hand, blue and green light, characterized by shorter wavelengths, become the dominant light in the underwater images. The right side of Fig. 1 visually demonstrates the color deviation in the original underwater image. This deviation is evident in the histogram of the red channel, which heavily leans towards lower pixel values. Conversely, the tricolor histogram of the enhanced image displays a more uniform distribution. Furthermore, the loss of light energy exacerbates underexposure, particularly affecting the contrast in regions far from the camera [5].

☆ This work was supported by National Natural Science Foundation of China (61972064), LiaoNing Revitalization Talents Program (XLYC1806006) and the Fundamental Research Funds for the Central Universities, China (DUT19RC(3)012).

* Corresponding author at: School of Innovation Experiment, Dalian University of Technology, Dalian 116081, Liaoning, China.
E-mail address: fenglin@dlut.edu.cn (L. Feng).

https://doi.org/10.1016/j.asoc.2024.112000
Received 6 March 2024; Received in revised form 28 June 2024; Accepted 11 July 2024
Available online 15 July 2024

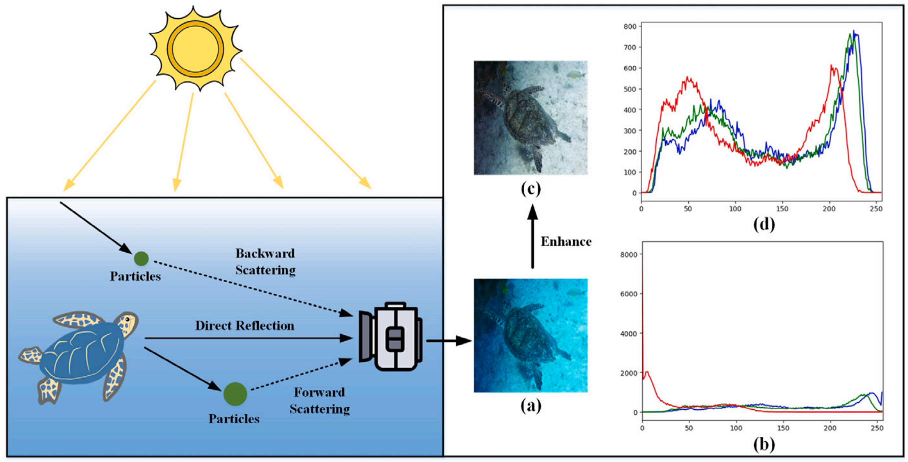

Fig. 1. Simple imaging process of underwater scene and comparison before and after enhancement, where (a) and (c) are the original underwater image and the enhanced image of our method, (b) and (d) are the tricolor histograms corresponding to the two images.

To enhance underwater visibility, artificial illumination is frequently employed to supplement the available light source. However, this method often fails to uniformly illuminate the underwater scene, exacerbating the challenges associated with underwater vision quality [6]. Consequently, the development of effective underwater image-processing methods has emerged as a prominent research focus in recent years. These methods can be broadly categorized into three groups: non-physical model methods, physical model methods, and deep learning methods. Non-physical model methods focus on the issue of color channel loss in underwater images. These methods achieve improved visibility of underwater images by altering pixel values in distinct regions. Physical model methods, on the other hand, establish imaging models rooted in the optical properties of the underwater environment. In contrast to haze images with similar imaging process and enhancement purpose, underwater imaging accounts for the wavelength-dependent effects of light propagating underwater. Among them, the Jaffe-McGlamery underwater optical model has received extensive attention in this domain and is often defined as follows [7]:

$$I = J \cdot e^{-\beta \cdot d} + B\left(1 - e^{-\beta \cdot d}\right), \tag{1}$$

where I and J represent the captured underwater image and the target restoration image, respectively, and B represents the background light. β represents the light attenuation coefficient, d represents the camera-shooting point distance, and $e^{-\beta \cdot d}$ represents the underwater transmission map. By deriving the degradation process, the physical model methods restore the underwater image [8].

In recent years, deep learning has received significant attention across various domains [9–15]. In contrast to traditional manual methods, deep learning methods harness extensive training data to learn intricate nonlinear mapping functions, thereby establishing potent capabilities for enhancing underwater images. Among these methods, the encoder–decoder architecture is frequently employed to capture multi-scale features. Many methods based on this architecture construct deep neural networks and utilize skip connections to address the issue of gradient degradation. As the network reconstructs features to higher resolutions, the pre-features extracted by the encoder are amalgamated with the post-features generated by the decoder. This mechanism reuses features across deep mappings. However, the pre- and post-features undergo significant alterations through multiple mappings, and the direct connection ignores differences in quality between the two. Consequently, during the connection process, blurred features have impacted clear features, resulting in suboptimal performance of the fused features in terms of spatial details. Furthermore, as the depth of the network architecture increases, the features endure multiple mapping stages, leading to information loss that affects the visibility of the generated images.

To address these challenges, we introduce a cross-transformer network focusing on pre-post differences, enabling the construction of a low-quality to high-quality enhancement process for underwater images. The main contributions of this paper are as follows:

1. We introduce a novel cross-wise transformer module (CTM), which employs cross-attention to capture dependencies among multi-scale features. CTM facilitates the fusion of pre-post features through interactive learning, enabling more accurate attention to detail-degraded regions.

2. We devise a feature supplement module (FSM) that employs an information complementary mechanism. FSM fuses features derived from images of different scales, thereby compensating for the information loss in deep networks.

The remainder of this paper is organized as follows: Section 2 presents a survey of related work in the realm of underwater image enhancement. Section 3 provides the detailed structure and loss function of the network. Section 4 conducts experimental analysis and performance discussion on multiple datasets. Finally, Section 5 concludes this paper.

2. Related work

In this section, we review related work in the field of underwater image enhancement, which can be mainly divided into three categories: non-physical model methods, physical model methods, and deep learning methods.

2.1. Non-physical model methods

Ancuti et al. [16] performed gamma correction and sharpening on the two images output by white balance, and then fused the two images based on a multi-scale fusion strategy. By constructing the exposedness-based weight maps, Zhu et al. [17] fused the image sequences with different exposures obtained by gamma correction with different power indices, and then linearly adjusted the saturation to enhance the images. Tao et al. [18] obtained two gain images with artificial multi-exposure fusion and local color correction, respectively, and the fusion of the two images was assisted by guided filtering and color channel transfer. Liang et al. [19] performed a piecewise linear transformation on each color channel according to the attenuation map, and then used a multi-scale decomposition method to remove the haze effect and compensate for the detail loss. Zhang et al. [20] first compensated the color channels using the designed attenuation matrix, then fused and sharpened the contrast images obtained by two improved histogram methods. Dong et al. [21] first measured the attenuation degree of the channel for reference compensation, then enhanced the local contrast and color in the LAB color space, and finally sharpened the details of the output results. Li et al. [22] identified image pixel colors through background light estimation, then applied two color compensation schemes to deal with color distortion, and finally applied haze-line theory to enhance blurred details. Considering the variability between different regions of underwater images, Zhou et al. [23] estimated the degree of drift by the statistical features of each region to guide the adaptive enhancement of degraded images. Zhang et al. [24] introduced a minimum color loss principle to adjust the image colors, and balanced the difference between the two channels to further enhance colors. Kang et al. [25] decomposed the two enhanced images into three independent components, then performed different improvements to fuse the enhanced images.

Non-physical model methods can enhance underwater image quality without necessitating prior knowledge. However, due to the intricate nature of the underwater environment, these methods may not always divide image regions in an optimal manner, which can lead to the introduction of unexpected color deviations or artifacts.

2.2. Physical model methods

Inspired by the adaptive retinal mechanism of teleost fish, Gao et al. [26] simulated the retinal feedback mechanism, color opponent mechanism, and center–surround opponent mechanisms, respectively, so as to enhance the color, detail, and contrast of underwater images. Berman et al. [27] extended the fog-line model [28] based on the wavelength-dependent attenuation of different water bodies, and compensated the transmission of different color channels. Marques et al. [6] used local contrast information to construct two atmospheric lighting models with different emphases, and generate the result by combining the restored images of the two models. Desai et al. [29] introduced backward scattering, direct scattering, and veiling light parameters into the Jaffe-McGlamery underwater optical model, and synthesized underwater-style images to train conditional generative adversarial network. Miao et al. [30] used an approximate single-viewpoint camera model to eliminate refraction distortion when light propagates underwater. Yuan et al. [31] first obtained the input for removing the fogging effect by dark channel prior, then gets the color correction input by compensating the lost color channel in CLELAB color space, and finally combines the two inputs to enhance the underwater texture. Li et al. [32] measured the influence of light attenuation and object distance on the transmission map when applying the color-line model, making the model more adaptable to the variable underwater environment. Zhuang et al. [33] introduced hyper-laplacian reflectance priors in the retinex model, effectively addressing blurry details and unnatural colors. Zhou et al. [1] introduced channel intensity prior in depth map prediction, and eliminated scattering effects by adaptive dark pixels.

Physical model methods aim to restore underwater images to their original appearance. Since predefined parameters exert a significant impact on performance, re-adaptation and complex calculations are necessary in different environments.

2.3. Deep learning methods

Jamadandi and Mudenagudi [34] introduced wavelet transform into the encoder/decoder structure and used high-quality underwater images to guide the style transfer of low-quality underwater images. Li et al. [35] used the underwater imaging model to synthesize underwater images with different degradation degrees, and effectively improved the quality of underwater images and underwater videos. Yang et al. [36] designed an adaptive feature fusion module to replace upsampling and downsampling, and significantly reduced the number of model parameters. Wang et al. [37] introduced HSV space into underwater image deep learning for the first time. The proposed network performs color deviation removal and saturation adjustment in RGB space and HSV space, respectively, and then uses attention maps to fuse the two spatial results. Li et al. [38] extracted and combined features from three color spaces, and used attention and the medium transmission guidance module to make the network focus on different regions, allowing for more comprehensive enhancement of underwater images. Ren et al. [39] cleverly integrated swin transformer into U-Net, enhancing the ability to capture global dependence while ensuring the establishment of local dependencies, ultimately generating high-quality results. Zhou et al. [40] used two traditional enhancement methods to enrich the feature inputs, and then designed a new fusion module and channel attention module to fuse and enhance the image features, respectively.

As pioneers of unsupervised underwater methods, [41,42] first introduced the generative adversarial network (GAN) into underwater scenes, and generated underwater style images for training the color correction network. Hambarde et al. [43] used coarse to fine GAN network to estimate the depth map, and thus enhanced the underwater images. Inspired by transfer learning, Jiang et al. [44] applied the domain adaptation framework to two-step underwater image enhancement. Han et al. [45] introduced a lightweight encoding and decoding architecture into the GAN model, and enhanced the global-local correlation by spatial attention and channel attention. Cong et al. [46] combined the advantages of physical model and GAN to enhance underwater images, and designed a dual-discriminator structure to judge the reconstruction results. Liu et al. [47] introduced underwater object detection and contrastive learning to guide image enhancement, thus avoiding the need for paired data. Li et al. [48] proposed a contrast learning framework for multi-reference learning and discriminated the quality of different regions.

The complex models created through deep learning methods significantly enhance the visual quality of underwater images. However, these deep networks often neglect the loss of information during the progressive mapping of features, resulting in local detail loss in the enhanced images.

3. Methodology

In this section, we first introduce an outline framework of the proposed network, then introduce the cross-wise transformer module designed in this paper, and finally introduce other modules and loss functions employed within the network.

3.1. Network architecture

The left side of Fig. 2 shows the specific structure of the proposed network, primarily composed of three encoders/decoders and two cross-wise transformer modules (CTM). During the encoding and decoding processes, we incorporate the feature supplement module (FSM), residual learning module (RLM), and efficient channel attention (ECA), in addition to the basic convolution/deconvolution operations and ReLU activation function. The detailed configurations of FSM, RLM, and ECA are provided in (a), (b), and (c) on the right side of Fig. 2, respectively, while CTM is shown in Fig. 3.

For a single original underwater image denoted as I, we first apply the nearest neighbor interpolation twice to execute two down-sampling operations. This process yields two distinct-scaled images, namely $I^{\frac{1}{2}}$ and $I^{\frac{1}{4}}$, which are used to enhance the feature inputs for the network. Subsequently, within the initial stages of the three encoders, we deploy successive convolutional and ReLU functions. These operations work progressively to extract features and expand channels for the three inputs. To ensure an optimal balance between performance and speed in the context of multi-layer convolution, we alternate between using 3×3 and 1×1 kernels for convolutional layers. The former facilitates effective feature extraction, while the latter aids in channel expansion. Following three rounds of cross-convolutions, we establish a connection

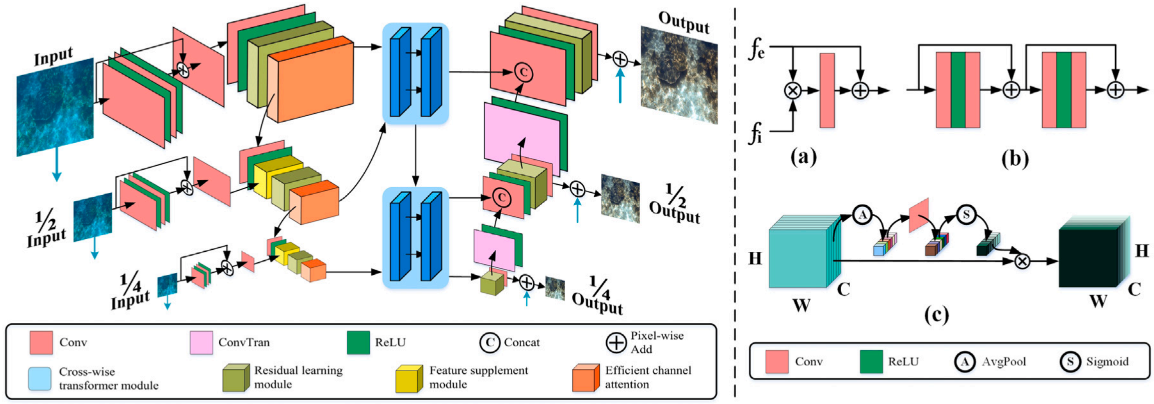

Fig. 2. Structure of the proposed model. The left side portrays the network structure, and the right side illustrates the FSM (a), RLM (b), and ECA (c). The network enhances underwater images across three scales, and inputs the original images at 1/2 and 1/4 scales as supplementary features. The final outputs at three scales are employed for loss calculation.

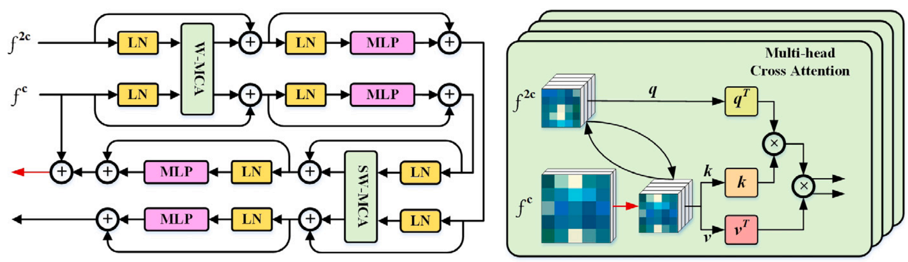

Fig. 3. Structure of CTM, where the two MCA are based on window partition and shift window partition, respectively. The structure of MCA is shown on the right side, with the red arrow representing the operation of changing scales.

between the expanded feature and the original input. This arrangement is designed to enhance the preservation of the original features. Additionally, we employ an additional 1 × 1 convolution to further refine the connected features. Consequently, we obtain three refined inputs with 32, 64, and 128 channels.

In contrast to the first encoder, the second and third encoders adopt the designed FSM to enhance the overall color and local details. FSM contains two types of inputs: encoded features f_e and refined input features f_i. The former are derived from the encoded features with the refined inputs and contain global context and semantic information. The latter, on the other hand, are the progressive expansion results of the original input, providing comprehensive image information. We down-sample f_e by a 3×3 convolution with a stride of 2 and add a nonlinear mapping by the ReLU function. After twice the scale reduction and channel expansion, the encoded features remain consistent with the refined input in terms of dimensions. Fig. 2(a) shows the fusion mechanism of the FSM, which employs an uncomplicated structure to ensure network efficiency. In the FSM, we perform an element-wise multiplication of the encoded features f_e with refined input features f_i. This operation acts as a feature selection mechanism, emphasizing common features in both inputs while suppressing irrelevant features such as noise and redundant information. This helps to enhances key features in the image, making them more prominent in subsequent network layers. Subsequently, we use 3 × 3 convolution to capture the complementary information between the two inputs, and combine it with f_e to obtain the output feature f_{fsm}. The procedure is as follows:

$$f_{fsm} = \varphi\left(f_e \otimes f_i\right) + f_e, \tag{2}$$

where $\varphi(\cdot)$ represents convolution, $\otimes$ represents element-wise product. Corresponding to the two-stage scales in the network, we employ features at 1/2 and 1/4 scales for FSM fusion. The 1/2 scale features retain relatively more detailed information, while the 1/4 scale features are more concerned with the global structure. Section 4.4 provides further experiments on the feature fusion under different scales. By leveraging the features extracted from the down-sampled images $I^{\frac{1}{2}}$ and $I^{\frac{1}{4}}$, the FSM facilitates the integration of complementary information from the original inputs. This mechanism not only enhances the expressiveness of the encoded features, but also compensates for information loss in deep network learning.

In deep networks, the residual learning can solve network degradation and foster the retention of original features. The RLM is shown in Fig. 2(b), employing a convolution-ReLU-convolution structure to extract features, where each convolutional layer uses a 3 × 3 kernel. To ensure network efficiency, RLM uses only two residual connections to complete the combination before and after learning. RLM serves to extract residual features, thus countering the potential decline in gradient information. Following the residual learning process, the various channels of the feature contain information of differing saliency. To adaptively select this information while maintaining efficiency, we employ ECA to learn dependencies among each channel. ECA, serving as a lightweight channel attention module, has less impact on network speed. The ECA outputs from the three encoders will be employed as inputs for both the two-stage CTM and the subsequent encoder, respectively.

The ECA results of the second and third encoders serve as inputs to the first CTM, yielding two outcomes. We output them to the second CTM and the third decoder, respectively, while the results of the second CTM are output to both the second and first decoders. Unlike traditional skip connections, CTM can capture the correlations between features in different periods, enabling the adaptive fusion of encoding and decoding features. The third decoder first introduces the CTM results to the RLM. Subsequently, these RLM outcomes contribute to subsequent decoding learning, generating the enhanced images at corresponding scales. Between different decoders, we use a deconvolution with a stride of 2 to change the scale and channel of the features, and merge and convolve them with the second CTM features. Through a 3 × 3 convolutional operation, the three decoders reconstruct RLM-learned features from multiple channels to 3, and then combined them with original underwater images of corresponding scales. Consequently, the network finally outputs enhanced images at three scales.

3.2. Cross-wise transformer module

In recent years, the transformer has attracted substantial attention from researchers due to its remarkable ability to establish remote dependencies. Some methods [49–51] incorporate the transformer to construct encoder/decoder structures, which enables the multi-scale feature information to undergo a more accurate global interaction process, resulting in notable performance improvement compared to convolutional networks. However, while many transformer-based methods have established relatively comprehensive global dependencies, the enhanced images often still exhibit detail degradation. This is because, in the multi-layer mappings of deep networks, the decoded features have significant changes compared to the encoded features, while most methods only use skip connection when combining the pre-post features. Due to the quality difference between the two features, there is a certain loss in the spatial details of the enhanced images [52]. Inspired by Swin Transformer [53], we designed a new architecture, CTM, between the encoder and decoder to establish an interactive learning mechanism among features in different periods. To capture remote dependencies, Swin Transformer performs interactions between sequence patches through a multi-head self-attention (MSA) mechanism. In contrast, CTM introduces a multi-head cross-attention mechanism (MCA). MCA not only considers information interactions within the same scale, but also enables cross-scale feature fusion and interactions. By focusing on the differences between pre-post features, CTM responds more accurately to detail-degraded regions, making the network perform well in capturing and enhancing detailed textures.

Fig. 3 shows the framework of CTM, which has as input two features f^C and f^{2C} with different scales $2H \times 2\ W \times C$ and $H \times W \times 2C$. Correspondingly, CTM contains two pathways for feature learning, each employing MCA to facilitate interactive learning between the pre-features f^C and post-features f^{2C}. To ensure the same scale, the CTM first performs the following down-sampling operation:

$$f^{C'} = L_{C\uparrow}\left(LN\left(f^C\right)_R\right), \tag{3}$$

where $(\cdot)_R$ represents the rearrange operation for scale and channel transformation, LN represents the layernorm layer, and $L_{C\uparrow}$ represents the linear layer for channel expansion. Notably, we opt to down-sample the higher scale feature f^C, rather than up-sample the lower scale f^{2C}. This is because CTM can capitalize on the richer information in f^C to compensate for the information loss in f^{2C}. Additionally, the lower-scale attention input significantly reduces the computational cost of the transformer, preserving the network efficiency. Next, we input the two features into the MCA and slice them into non-overlapping windows. The number of windows is set to $P \times P$, and the size of the windows is $\frac{H}{P} \times \frac{W}{P}$. Through the linear layer, we obtain query matrices Q and Q', key matrices K and K' and value matrices V and V', which have a shape of $P^2 \times C$. These matrices are calculated as follows:

$$\begin{aligned} MCA &= \varepsilon\left(\frac{QK'^T}{\sqrt{d}} + b\right)V', \\ MCA' &= \varepsilon\left(\frac{Q'K^T}{\sqrt{d}} + b\right)V, \end{aligned} \tag{4}$$

where d represents the matrix dimension, $\varepsilon(\cdot)$ represents the softmax function, and b represents the learnable relative position deviation with a shape of $(2p-1) \times (2p-1) \times \frac{C}{P}$. Through cross-attention learning of the two features, the difference between pre-post features is reduced, contributing to a smoother fusion between the encoder and decoder. Moreover, CTM contains MCA based on two window partition methods: window partition (W-) and shift window partition (SW-). Unlike W-, SW- changes the pixel distribution of features to achieve varying window partitions. The alternation between these two partition methods extends the remote modeling capability while interacting with different features [53].

According to the three-stage features f^C, f^{2C}, and f^{4C} of the network, we establish two sequential stages of CTM. At the end of the first CTM, the learning features associated with f^C are first up-sampled to the original input scale. Subsequently, these features are fused with the original input features, thereby preserving more intricate details. The CTM-learned features correspond to f^C and f^{2C} are output to the third decoder and the next stage, respectively. In the second CTM, another input originates from the ECA result f^{4C} of the third encoder. The learning features associated with f^{2C} and f^{4C} are both fused with the original input at the end, and then output to the decoders for the next fusion. The introduction of CTM allows the network to capture feature dependencies across varying scales, thus paying more attention to the differences between the pre-post features. When the network restores the features from low scale to input scale, CTM protects detail information by the interaction of features from different stages, resulting in enhanced underwater images with improved detail and contrast.

3.3. Efficient channel attention

The features extracted by the three encoders have channels with different information, signifying varying degrees of channel saliency. Therefore, we employ ECA [54] to establish dependencies among the feature channels. The specific flow of ECA is illustrated in Fig. 2(c). Unlike SE attention [55], ECA does not perform channel compression on the pooling result $p^{C\times1\times1}$ after the avgpool operation on the input features $f^{C\times H\times W}$, but establishes inter-channel dependencies by a 1-dimensional convolution. The obtained weights, derived from the sigmoid function, are then multiplied with the original input to obtain the output features f_o:

$$f_o = f \otimes \left(\varphi_1(p)\right)_{\mathrm{Sig}}, \tag{5}$$

where $(\cdot)_{Sig}$ denotes the sigmoid function, and $\varphi_1(\cdot)$ represents the 1-dimensional convolution. The results of three ECA are input to the CTM for the next learning. By integrating channel-focused ECA and spatial-focused CTM, the network can effectively establish the dependencies of features across both channel and space. This dual-focus contributes to the network in comprehensively improving underwater images.

3.4. Loss function

With the aid of pre-trained deep networks, the perceptual loss extracts high-level features and quantifies the differences between them. This is beneficial for addressing image blurring and color distortion caused by underwater degradation, ultimately optimizing image quality. Compared to other loss functions, the perceptual loss not only focuses on pixel-level differences, but also captures feature similarities between images. With this property, the training phase can encourage the network to perform better in terms of clarity and color, resulting in images that are more in line with human visual perception.

In this paper, we first apply the perceptual loss L_p based on human eye perception. By pre-training the VGG-19 network [56] on the ImageNet dataset [57], the image features are extracted and used to quantify the feature similarity between images. The convolutional layer denoted as $\mathbb{C}_i(\cdot)$, corresponds to the ith layer of the network. The perceptual loss between two images is defined as follows:

$$L_p = \left|\mathbb{C}_i(x) - \mathbb{C}_i(y)\right| \tag{6}$$

In addition, this paper applies the edge loss L_e to motivate the network to generate edge-clear results. This loss is calculated as follows:

$$L_e = \sqrt{\|\ell(x) - \ell(y)\|^2 + \beta^2}, \tag{7}$$

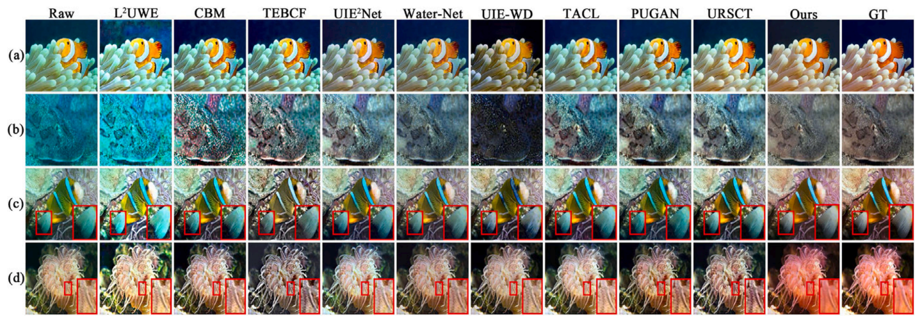

Fig. 4. Visual comparison of all methods on the CycleGAN-600 dataset, where the Raw and GT represent the original underwater images and ground truth.

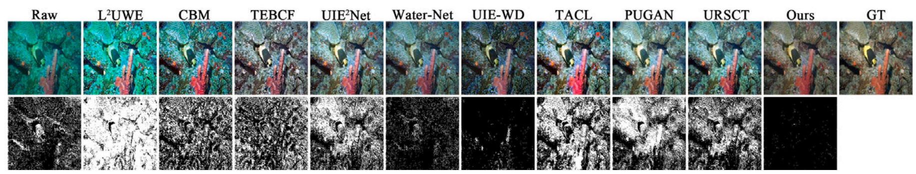

Fig. 5. Residual comparison of all method results. The wider the white region, the greater the difference with ground truth.

where $\ell(\cdot)$ represents the Laplacian operator, β is a constant for the stable loss value, which is set to 10^{-3}. The output of the network consists of three images with different scales. In this paper, the loss L^1, $L^{\frac{1}{2}}$, and $L^{\frac{1}{4}}$ of each image are obtained by combining two losses, and the total loss L_{total} of the training is obtained by superimposing the losses of three scales:

$$L^i = \lambda L_p\left(x^i, y^i\right) + L_e\left(x^i, y^i\right), \tag{8}$$

$$L_{\text{total}} = L^1 + L^{\frac{1}{2}} + L^{\frac{1}{4}}, \tag{9}$$

where λ is the balance factor set to 0.05 to balance the two losses. When the total loss between the three results and the corresponding ground truth decreases, the model will exhibit better enhancement performance.

4. Experiment

In this section, we firstly describe the experimental details, secondly compare the performance of the proposed network with other advanced methods on different datasets, thirdly conduct the ablation study to validate the effectiveness of the module, fourthly perform the extended application on other visual tasks and poor-lighting scenarios, and finally explore the complexity and limitations of the model.

4.1. Implementation details

Compared methods: The method was compared with nine underwater advanced methods, including three traditional methods: (L^2UWE [6] based on local contrast and multi-scale fusion, CBM [58] based on contour bougie morphology, and TEBCF [31] based on blurriness and color fusion), six deep learning methods: (Water-Net [59] using gated fusion network, UIE^{2}Net [37] based on two color spaces, and UIE-WD [60] using wavelet-based dual-stream network, TACL [47] based on twin adversarial contrastive learning, PUGAN[46] based on physical model and GAN, and URSCT [39] based on U-Net and swin transformer).

Experimental datasets: In the recent field of underwater image enhancement, many related researches generally adopt two solutions: X→Y or Y→X, in producing the datasets required for deep learning techniques. One is to enhance low-quality underwater images into high-quality underwater images by various advanced methods. Conversely, the other is to transform high-quality images into lower quality images by simulating the multi-stage degradation processes. In addition, GAN is prevalent for generating paired datasets of two solutions. To train our model, we used 6128 pairs of underwater/normal images from the CycleGAN dataset [42], from which we randomly selected 600 pairs for reference test and the remaining 5528 pairs for training and validation. Moreover, we used the ImageNet dataset [57] (1813 images), the UFO dataset [61] (120 images), the EUVP dataset [62] (100 images), and the UIEB dataset [59] (60 images) for the no-reference test, which contain a variety of underwater scenes and different degrees of degradation.

Table 1
Quantitative comparison of all methods on the CycleGAN-600 dataset (**optimal**, sub-optimal).

	LPIPS	PSNR	SSIM	FSIM	UICM	UIQM
L^2UWE	0.4141	14.920	0.6483	0.7959	−78.022	2.5712
CBM	0.3138	21.443	0.7420	0.8502	−31.691	4.0319
TEBCF	0.3172	18.774	0.6862	0.8130	−14.054	4.5428
Water-Net	0.2583	22.884	0.8107	0.9029	−17.386	4.5038
UIE^{2}Net	0.2698	21.570	0.7889	0.8971	−17.384	4.5349
UIE-WD	0.3194	17.439	0.6989	0.8615	−7.2043	4.7096
TACL	0.2721	21.586	0.7628	0.8812	−23.336	4.4296
PUGAN	0.2867	21.365	0.7808	0.8843	−20.268	4.4017
URSCT	0.2684	21.051	0.7863	0.8895	−20.454	4.3138
Ours	**0.1605**	**25.974**	**0.8571**	**0.9258**	**−1.4521**	**4.9790**

Evaluation metrics: To objectively evaluate the enhancement effect, we used four reference evaluation metrics: perceptual similarity (LPIPS) [63], PSNR, SSIM, feature similarity (FSIM) [64], and non-reference underwater quality assessment metrics UICM and UIQM [65].

Environment setting: The proposed model was built with the PyTorch library and experimented on a server with Ubuntu 18.04 OS and TITAN RTX graphics card. We set the initial learning rate and the total batches to 0.0001 and 100, and used the Adam optimizer to adjust the learning rate batch by batch.

4.2. Comparison on reference dataset

Figs. 4 and 5 provide visual and residual comparisons of the results generated by all methods on the CycleGAN-600 dataset. L^2UWE improved the sharpness of the image, but the color deviation of the original image was severely deepened. As a result, the residual image between the enhanced image and the ground truth had the densest white region. CBM failed to remove the effect of underwater color

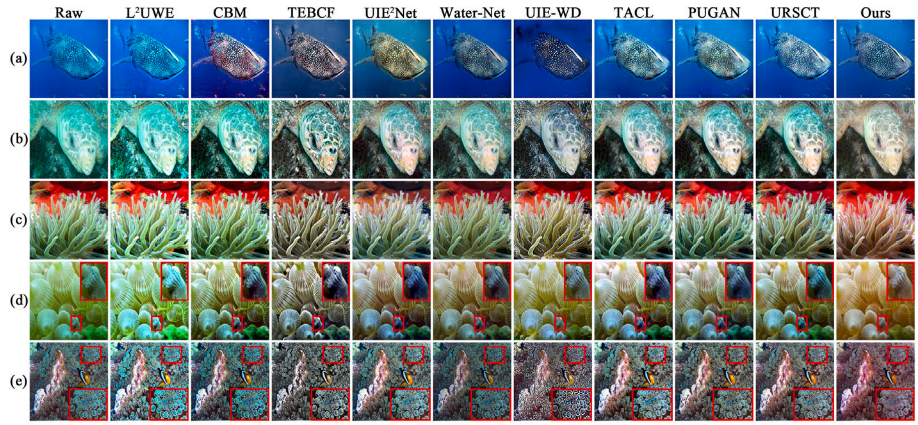

Fig. 6. Visual comparison of all methods on the Imagenet-1813 dataset.

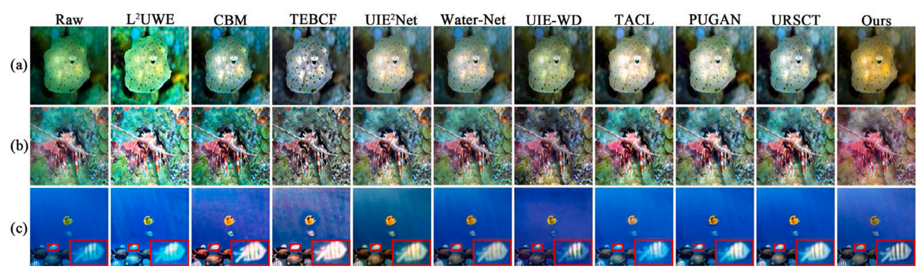

Fig. 7. Visual comparison of all methods on the UFO-120 dataset.

deviation, resulting in an amplification of noise in the raw underwater image. On the other hand, TEBCF enhanced the contrast of the original input, but the overall color is dull. The enhanced results of TACL and PUGAN exhibited considerable color deviation. In contrast, UIE^{2}Net and URSCT mitigated the impact of color deviation, but the enlarged details underscored deficiencies in color and saturation. Consequently, compared to the original image, the residual images of these seven methods exhibited more extensive white regions. Color deviation remained in Water-Net, negatively impacting clarity and contrast. The results of UIE-WD exhibit rich details but with lower brightness. Compared with the above methods, our method not only completely removed the influence of color deviation, but also had the most prominent vivid colors and detailed textures. Notably, the smallest white region in the residual image illustrates that our results are closest to the ground truth. Table 1 shows the quantitative metrics for all methods, and our method obtained the best values across all metrics, further demonstrating the good performance of color and clarity.

4.3. Comparison on non-reference datasets

Further visual comparisons were conducted across other methods on four datasets: ImageNet-1813, UFO-120, EUVP-100, and UIEB-60. The corresponding results are illustrated in Figs. 6- 9, respectively.

Fig. 6 show the results of the different methods on the ImageNet-1813 dataset. L^{2}UWE exacerbates color deviations of the original image. CBM and TEBCF enhance sharpness and contrast in underwater scenes, but also exhibit over-enhancement of the red channel and blurring of details in darker regions. Some results of UIE^{2}Net are affected by the yellow color, which destroys the original appearance. Results of UIE-WD have substantial noise in detail-rich regions, rendering some regions densely black. Water-Net, TACL, PUGAN, and URSCT significantly improved image contrast, but enlarged details exhibited low brightness and color deviation. As can be seen in (b) and (c), our proposed method excels in rectifying underwater color deviations, restoring the original appearance of the image. The enhanced results exhibit attractive color and contrast, and the enlarged details exhibit excellent visibility.

In Figs. 7, 8 and 9, the results of L^{2}UWE exhibit overexposure, which consequently affect the texture contrast in the images. TEBCF, UIE-WD, and Water-Net display low brightness, with UIE-WD also exhibiting unnatural artifacts. The enlarged details of the CBM and TEBCF highlight red dominance and overexposure. UIE^{2}Net, TACL, and PUGAN significantly enhanced colors, but do not produce ideal results in scenes characterized by chaotic colors. URSCT details exhibit good contrast, highlighting sharper texture features. In the majority of cases, the results of most methods still exhibit notable color deviations. Our results exhibit the purest colors and are least affected by other colors and artifacts, with correspondingly sharper textures in enlarged details.

Table 2 shows the superiority of our method, primarily due to its remarkable color and contrast, positioning it ahead of other methods. However, it is noteworthy that the smoothness of our method on the EUVP-100 dataset affects the UIQM metric.

4.4. Ablation study

On ImageNet-1813 and CycleGAN-600 datasets, we conducted an ablation study on networks with different configurations to demonstrate the effectiveness of each module. The configurations are as follows: (1) -w/o FSM: without feature supplement module, (2) -w/o RLM: without residual learning module, (3) -w/o ECA: without efficient channel attention, (4) -w/o CTM: without cross-wise transformer module, (5) -r/w ST: replace CTM with Swin Transformer.

Fig. 10 and Table 3 show the visual and metric comparisons for networks with different configurations, respectively. The absence of FSM and RLM impacts the color and contrast of images to a certain degree, thereby influencing the corresponding metrics. The -w/o ECA, due to the lack of ECA for channel importance identification, results in less vibrant color and performs poorly in underwater clarity. In

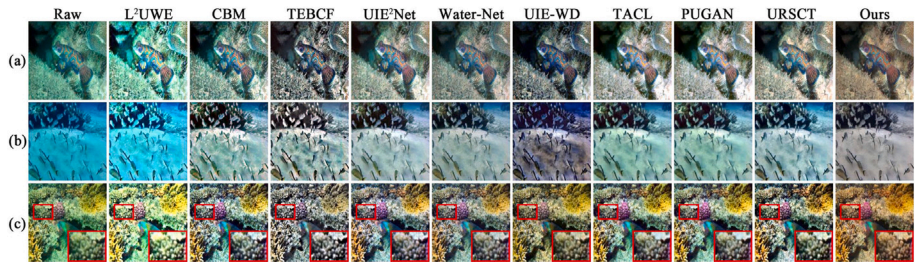

Fig. 8. Visual comparison of all methods on the EUVP-100 dataset.

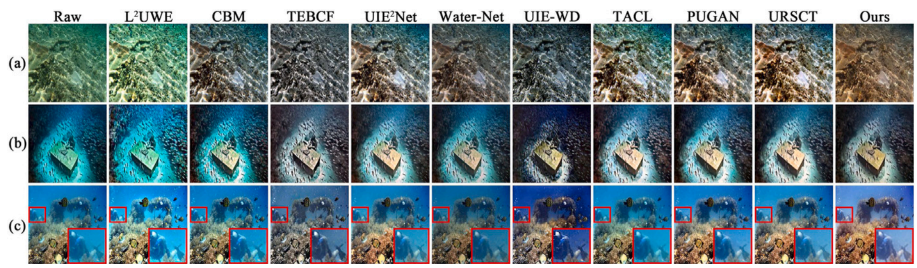

Fig. 9. Visual comparison of all methods on the UIEB-60 dataset.

Table 2
Quantitative comparison of all methods on the four non-reference datasets (**optimal**, sub-optimal).

	ImageNet-1813		UFO-120		EUVP-100		UIEB-60	
	UICM	UIQM	UICM	UIQM	UICM	UIQM	UICM	UIQM
L^2UWE	−71.411	2.8671	−68.978	2.8000	−76.377	2.7067	−58.638	4.5867
CBM	−32.482	4.2152	−34.936	3.8497	−30.935	4.0219	−32.513	4.0079
TEBCF	−13.171	4.5929	−20.163	4.2625	−15.355	4.4515	−14.161	4.4613
UIE^{2}Net	−17.395	4.6671	−22.073	4.2867	−18.790	4.5242	−19.281	4.5465
Water-Net	−18.990	4.5979	−21.887	4.2835	−21.199	4.4300	−18.304	4.5579
UIE-WD	−3.6991	4.1167	−3.3134	4.7993	−5.8213	4.7271	1.4814	4.8945
TACL	−20.416	4.5767	−33.616	4.0472	−26.012	4.3499	−19.802	4.4576
PUGAN	−18.885	4.6635	−26.494	4.1314	−21.477	4.3713	−19.559	4.5275
URSCT	−19.189	4.5260	−26.398	4.0611	−21.081	**5.6616**	−16.984	4.5266
Ours	**2.8190**	**5.6165**	**3.4094**	**4.8883**	**0.8768**	5.1706	**15.4230**	**5.0824**

Table 3
Quantitative comparison of all configurations on the ImageNet-1813 and CycleGAN-600 datasets (**optimal**, sub-optimal).

	CycleGAN						ImageNet		
	LPIPS	PSNR	SSIM	FSIM	UICM	UIQM	UICM	UIQM	Time (s)
-w/o FSM	0.1672	25.791	0.8505	0.9218	−4.524	4.8021	−2.5284	5.0523	0.0616
-w/o RLM	0.1748	25.735	0.8504	0.9203	**−0.5453**	4.8937	1.9351	4.9751	0.0533
-w/o ECA	0.1786	25.893	**0.8738**	0.9250	−1.1548	4.8704	0.2144	5.2761	0.0583
-w/o CTM	0.1610	25.822	0.8557	0.9246	−1.0366	4.9064	**4.7229**	5.1321	**0.0397**
-w/o ST	**0.1539**	25.390	0.8353	0.9255	−3.1029	4.9033	−0.7719	4.9622	0.0749
Ours	0.1605	**25.974**	0.8571	**0.9258**	−1.4521	**4.9790**	2.8190	**5.6165**	0.0755

the enlarged detail figure (e), compared to other configurations, the texture of -w/o CTM is more blurred, and the color deviation of -r/w ST is more severe. Additionally, -r/w ST has the largest white region in Fig. 11, indicating the greatest deviation from the ground truth. Table 3 demonstrates that the -w/o CTM has the highest efficiency, but the poor performance of the underwater contrast adversely affects the UIQM score.

The experimental results show that in the absence of CTM, the details are not well protected due to the ignored differences between pre-post features. In contrast, the addition of CTM leads to clearer and more prominent image details, ultimately enabling the network to generate images with improved contrast. Notably, whether in visual or residual comparisons, the results of the complete network exhibit better visual effects and consistency with the ground truth, thereby obtaining the optimal and sub-optimal metrics. Considering runtime, CTM does influence network efficiency to a certain extent, attributable to the computation of positional correlation calculation in the two attention layers. Although the network increases the computational cost due to various components, it still maintains a good processing speed.

In addition, we also conducted an ablation study on different scales of FSM fusion features, namely {1/8, 1/16}, {1/4, 1/8}, {1/2, 1/4}. Fig. 12 shows the enhancement results by combining features under different scales. It can be seen that the network using the features under {1/8, 1/16} scales exhibits a large number of color patches, while {1/4, 1/8} suffers from obvious color deviation. In contrast, the network using the features under {1/2, 1/4} scales performs better in both color accuracy and noise resistance, achieving essentially the optimal and sub-optimal values in Table 4. Therefore, we use the features under the {1/2, 1/4} scales as inputs to better compensate for information loss in the network.

Table 4
Quantitative comparison of different scale features on the ImageNet-1813 and CycleGAN-600 datasets (**optimal**, sub-optimal).

	CycleGAN						ImageNet		
	LPIPS	PSNR	SSIM	FSIM	UICM	UIQM	UICM	UIQM	Time (s)
{1/8, 1/16}	0.1865	25.035	0.8253	0.9044	**−0.3874**	4.8579	−0.1845	5.0531	0.0579
{1/4, 1/8}	**0.1579**	25.959	0.8470	0.9218	−3.8735	4.8115	0.1223	5.7604	0.0661
{1/2, 1/4}	0.1605	**25.974**	**0.8571**	**0.9258**	−1.4521	**4.9790**	**2.8190**	5.6165	0.0755

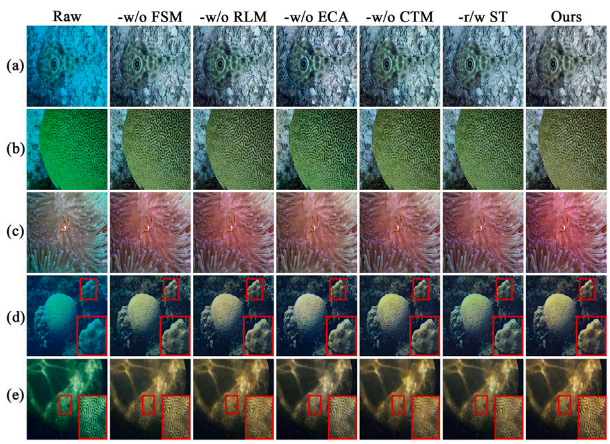

Fig. 10. Visual comparison of all configurations on the ImageNet-1813 datasets.

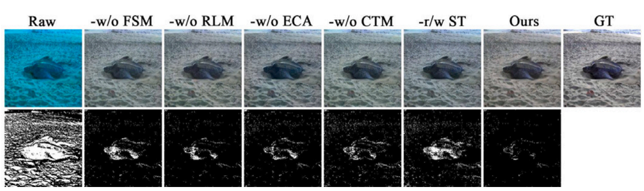

Fig. 11. Residual comparison of all configurations on the CycleGAN-600 dataset.

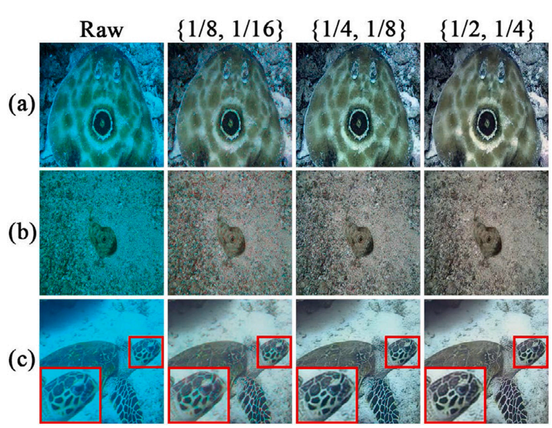

Fig. 12. Visual comparison of different scale features on the CycleGAN-600 dataset.

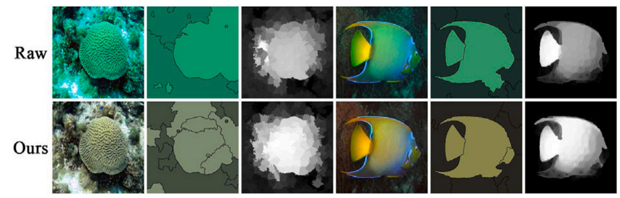

Fig. 13. Enhancement effect of the proposed method in visual tasks.

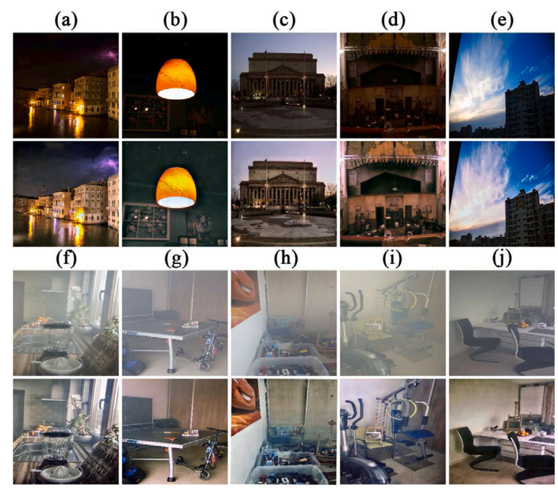

Fig. 14. Application of the proposed method to low-light and overexposure scenes.

4.5. Cross-library experiment

We validated the model feasibility by two challenging tasks: image segmentation [66] and saliency detection [67], without any parameter tuning. As depicted in Fig. 13, the enhanced images exhibit more precise segmentation details and more prominent saliency features. Notably, we recognize that the variability of natural light conditions may have an impact on model performance. Therefore, we specifically tested the model under low-light scenes [68,69] and hazy scenes [70], which often face challenges such as poor visibility and reduced clarity [71,72]. As shown in Fig. 14, even under these challenging lighting conditions, our method still infuses images with more vivid color, effectively enhancing details, saturation, and contrast. This indicates that the model has good generalization ability and enhancement effect, firmly supporting its practical applicability.

4.6. Discussion

We compared the complexity of all models, Table 5 shows FLOPs, parameters and runtime for all methods. It can be seen that the proposed method does not achieve the optimal space and efficiency. The model incorporates a three-stage enhancement process and interactive fusion of pre- and post-features, aiming to capture more contextual information and detailed features. While these designs consume some computing resources, the model demonstrates better performance.

In addition, we show some failure results of the proposed method in Fig. 15. Our method does not work well in scenes with severely degraded quality, such as turbid imaging or uneven lighting, etc. The issues can be analyzed as follows: Firstly, detailed information such as edges, textures, and shapes are severely lost during the imaging process in turbid water. The enhancement model cannot regenerate this missing information, so the clarity of the result is not attractive. Secondly, uneven lighting leads to brightness differences between image regions. Due to the lack of light sensitivity in our method, over-enhancement of the red channel occurs in regions with relatively high brightness.

Table 5
FLOPs, Parameters and Runtime of all methods (first, second, third).

Methods	FLOPs (G)	Parameters (M)	Runtime (s)
L²UWE	–	–	2.8814
CBM	–	–	0.2118
TEBCF	–	–	1.4066
Water-Net	142.9	109.1	1.8350
UIE²Net	26.06	0.535	0.9773
UIE-WD	12.84	13.70	1.0402
TACL	56.86	11.38	0.0767
PUGAN	0.676	0.010	0.0672
URSCT	18.11	11.26	0.0676
Ours	45.97	4.783	0.0755

Fig. 15. Failure results of the proposed method in severe turbid and uneven lighting scenes.

5. Conclusion

In this work, we propose an underwater image enhancement network focusing on pre-post differences. The network learns multi-scale features through the encoder–decoder structure and integrates FSM for fusing input features at varying scales. Different from similar methods, we do not simply connect the pre-post features directly, but first employ ECA to capture the dependencies among feature channels, and then incorporate CTM to diminish differences between the pre-post features. Ultimately, the network generates enhanced images across three scales, facilitating a comprehensive supervised process. The results on multiple underwater datasets and vision tasks show that our method achieves exceptional enhancement capabilities and generalization. Furthermore, the network components contribute diverse advantages for improved performance.

However, our method still requires improvement in terms of model complexity and computational efficiency, and its performance needs to be further improved when tackling severely degraded underwater scenes. In future work, we will focus on optimizing the model parameters and build a more lightweight architecture for efficient underwater video enhancement. Meanwhile, for the complex underwater environment, we intend to introduce physical cues or other modal information to facilitate more accurate enhancement. Moreover, we aim to explore an integrated framework that adapts to various poor-lighting conditions, with a view to leveraging its value in a wider range of applications.

CRediT authorship contribution statement

Zhixiong Huang: Writing – original draft, Methodology, Conceptualization. **Jinjiang Li:** Supervision, Methodology, Data curation. **Xinying Wang:** Writing – original draft, Formal analysis. **Zhen Hua:** Validation, Supervision. **Shenglan Liu:** Writing – review & editing, Investigation. **Lin Feng:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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